MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 10548133

(DETECTOR, PHASE)

1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of MIL-C-50652 comprise the complete requirements for the phase detector assembly used in the Range Computer CP-888/VPS-2, part of Radar Set AN/VPS-2.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-50652	Circuit Card Assemblies, General Specification for

DRAWINGS

U.S. Army, Frankford Arsenal

F10548133	Detector, Phase

PACKAGING DATA SHEET

10548133	Packaging of Phase Detector

FSC 1285

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:32Z
Check the source to verify that this is the current version before use.

(Copies of specifications, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. - The assembly shall be manufactured in accordance with Drawing F10548133 and drawings pertaining thereto.

3.2 General specifications. - The assembly shall meet the requirements of MIL-C-50652.

3.3 Environmental service conditions. - As specified in MIL-C-50652.

3.4 Electrical. - Each conductor shall be electrically continuous as specified on the schematic diagram of the phase detector assembly.

3.5 Performance.

3.5.1 Input voltages. - The phase detectors shall be supplied with the input voltages as specified in Table I for the requirements of 3.5.2 through 3.5.4.

TABLE I. - INPUT VOLTAGES

CHARACTERISTIC	TERMINALS
+15.0 ±0.1 Vdc, regulated; with an ac ripple less than 50 mV peak-to-peak.	P1-36, P1-37
-15.0 ±0.1 Vdc - same characteristics as +15.0 V.	P1-20, P1-21
+5.0 ±0.3 Vdc, regulated; with an ac ripple less than 50 mV peak-to-peak.	P1-30, P1-31, P1-32
+15, -15, +5 Vdc returns.	P1-22, P1-23, P1-24, P1-25, P1-26, P1-35, P1-36

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:32Z
Check the source to verify that this is the current version before use.

3.5.2 Sum and difference comparators. - With a 5 kilohertz (kHz) triangular wave of 1.0 volt (V) peak-to-peak, symmetrical about ground, and a 1.5 volts direct current (Vdc) level applied sequentially to the terminals as specified in Table II, there shall be a square wave at the output terminals specified in Table II. The square wave shall have a phase relationship to the input signal as specified in Table II with direct current (dc) levels between 0.0 ±0.5 V and +3.5 ±1.0 V.

TABLE II. - SUM AND DIFFERENCE WAVEFORMS

	SUM INPUT	DIFFERENCE INPUT	Re OUTPUT
STEP	P1-43[1/]	P1-47[1/]	P1-17[1/]
1.	5 kHz triangular wave 1.0 V peak-to-peak.	+1.5 Vdc	Square wave 180° out of phase with input.
2.	5 kHz triangular wave 1.0 V peak-to-peak.	-1.5 Vdc	Square wave in phase with input.
3.	+1.5 Vdc	5 kHz triangular wave 1.0 V peak-to-peak.	Square wave 180° phase with input.
4.	-1.5 Vdc	5 kHz triangular wave 1.0 V peak-to-peak.	Square wave

1/ Returns for sum input, and difference input signals are P1-41 and P1-45, respectively.

3.5.3 Sum signal strength response.

3.5.3.1 ECM relay driver signal output. - The ECM relay driver signal at P1-12 (return side of signal at P1-10) shall be as specified in Table III with the following conditions:

a. A 0.1 hertz (Hz) triangular wave, symmetrical about ground, with the amplitudes as specified in Table III applied to P1-49 (return side of signal at P1-51).

b. Logic level inputs as specified in Table III, with a logic 0 being a 0.0 ±0.5 Vdc level and a logic 1 being a +3.5 ±1.0 Vdc level.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:32Z
Check the source to verify that this is the current version before use.

c. The ECM output terminal P1-12 shall be connected to +5.0 Vdc in series with a 1 kilohm, 1/2 watt (W) resistor.

TABLE III. - INPUT-OUTPUT SIGNALS

	INPUT SIGNALS		OUTPUT SIGNAL
STEP	P1-14	P1-49	P1-12
1.	Logic 1	0.74 ±0.10 V peak-to-peak.	Adjustable by R20 so that when signal level at P1-49 is exceeded output switches from a logic 1 level to a logic 0 level.
2.	Logic 0	2.0 ±0.1 V peak-to-peak.	Output signal remains at logic 1 regardless of signal level at P1-49.

3.5.3.2 Target signal output. - The target signal output at P1-9 (return side of signal at P1-8) shall be adjustable by R22 to ensure switching from a logic 0 level to a logic 1 level when the amplitude of the applied 0.1 Hz input signal, as specified in 3.5.3.1, exceeds 0.5 ±0.1 V peak-to-peak.

3.5.4 Track relay driver signal. - The track relay driver signal at P1-14 shall be as specified in Table IV, when connected to +5.0 Vdc in series with a kilohm (K), 1/2 W resistor, and logic level input applied as specified in Table IV.

TABLE IV. - TRACK RELAY DRIVER SIGNAL

	SIGNAL INPUT	SIGNAL OUTPUT
STEP	P1-39[2/]	P1-13[2/]
1.	Logic 0	Logic 1
2.	Logic 1	Logic 0

2/ Return side of input and output signals are P1-40 and P1-11, respectively.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:32Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. - The quality assurance provisions of MIL-C-50652 shall apply to the assembly covered by this detail specification.

4.2 First article inspection. - As specified in MIL-C-50652.

4.3 Inspection provisions. - As specified in MIL-C-50652.

4.4 Special sampling. - As specified in MIL-C-50652.

4.5 Inspection equipment. - As specified in MIL-C-50652 and 4.5.2.3.

4.5.1 Government-furnished inspection equipment. - As specified in MIL-C-50652.

4.5.2 Contractor-furnished inspection equipment. - As specified in MIL-C-50652.

4.5.3 Test equipment. - In conjunction with 4.5 the following test equipment listed in Table V shall be utilized in the performance of the applicable tests as specified in 4.6.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:32Z
Check the source to verify that this is the current version before use.

TABLE V. - TEST EQUIPMENT LIST

ITEM NUMBER	DESCRIPTION
1. Power supply	5.0 ±0.3 Vdc, regulated. Ac ripple: less than 50 mV peak-to-peak. Capacity: 200 mA.
2. Power supply	15 ±0.1 Vdc, regulated. Ac ripple: less than 50 mV peak-to-peak. Capacity: 200 mA.
3. Power supply	Same as item 2.
4. Function generator	Triangular output waveform; frequency 0.1 Hz. Amplitude 3.0 V peak-to-peak or greater, into 50 ohms; signal symmetrical about ground.
5. Oscilloscope	Sweep period 100 μs/cm or greater. Input resistance: greater than 9 megohms.
6. Digital voltmeter	Visual display 3 digits or more, full scale range 20.0 V or greater. Input resistance: 1 megohm or greater. Accuracy: ±0.01%.
7. Test fixture	See Figure 1. See Table VI for test fixture parts list.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:32Z
Check the source to verify that this is the current version before use.

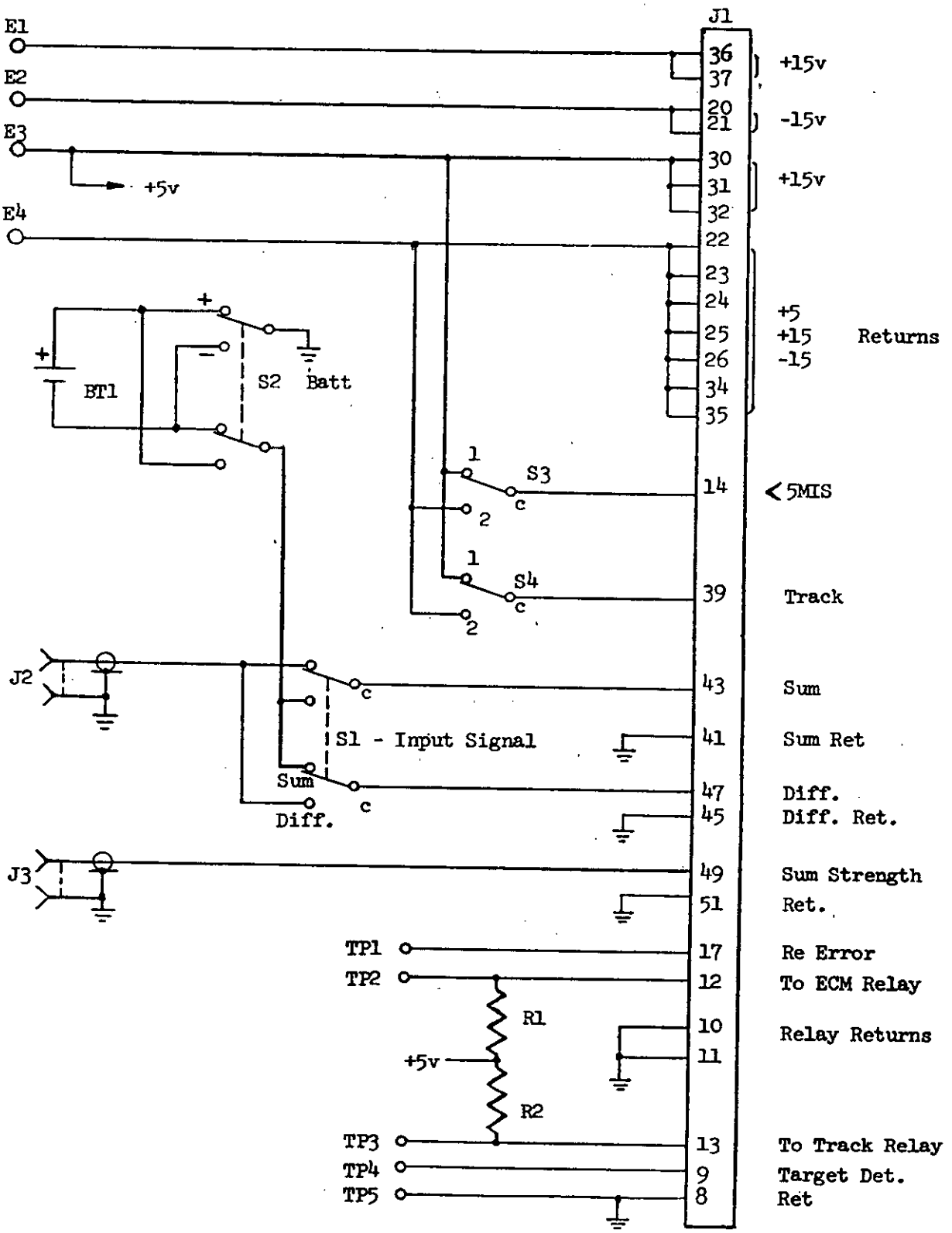


FIGURE 1. TEST FIXTURE SETUP.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:32Z
Check the source to verify that this is the current version before use.

TABLE VI. - TEST FIXTURE PARTS LIST

DESIGNATION	NOMENCLATURE AND DESCRIPTIONS
BT1	Battery, dry cell - 1.5 V.
E1 thru E3	Binding post, H.H. Smith 257-R or equivalent.
E4	Binding post, H.H. Smith 257-B or equivalent.
J1	Connector; any of the following or equivalent: (1) 5893232-1, Malco Mfg. Co., Inc. Chicago, Illinois. (2) 500946-01, National Connector Corp. Minneapolis, Minn. (3) 07-5422-0198-01, Elco Corp. Willow Grove, Pa.
J2	Connector, receptacle, UG-25B/U or equivalent.
J3	Same as J2.
R1 and R2	Resistor, 1K ohm, 1/2 W, 1.0%.
S1 and S2	Switch SPDT.
S3 and S4	Switch DPDT.
TP1 thru TP5	Tip jack, H.H. Smith 1501 or equivalent.
Z1 and Z2	Integrated circuit, SG-140-03, Sylvania or equivalent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:32Z
Check the source to verify that this is the current version before use.

4.6 Test methods and procedures.

4.6.1 Environmental. - As specified in MIL-C-50652.

4.6.2 Electrical. - Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.6.3 Performance. - The selected phase detectors shall be tested as follows: Set up equipment as shown in Figure 2.

4.6.3.1 Input voltages. - Use items 1 through 3 to supply input voltages as specified in Table I. Apply input voltages to E1 through E4 on the test fixture, item 7.

4.6.3.2 Sum and difference comparators. - Use function generator, item 4, to supply input signals as specified in 3.5.2. Apply signal to J2 and use switch S1 on test fixture, item 7 to switch input signal for either sum or difference inputs. Switch S2 on test fixture is used to switch the 1.5 Vdc levels for either positive or negative polarities as specified in Table II. At TP1 on the test fixture, item 7, observe the Re output signal as specified in Table II.

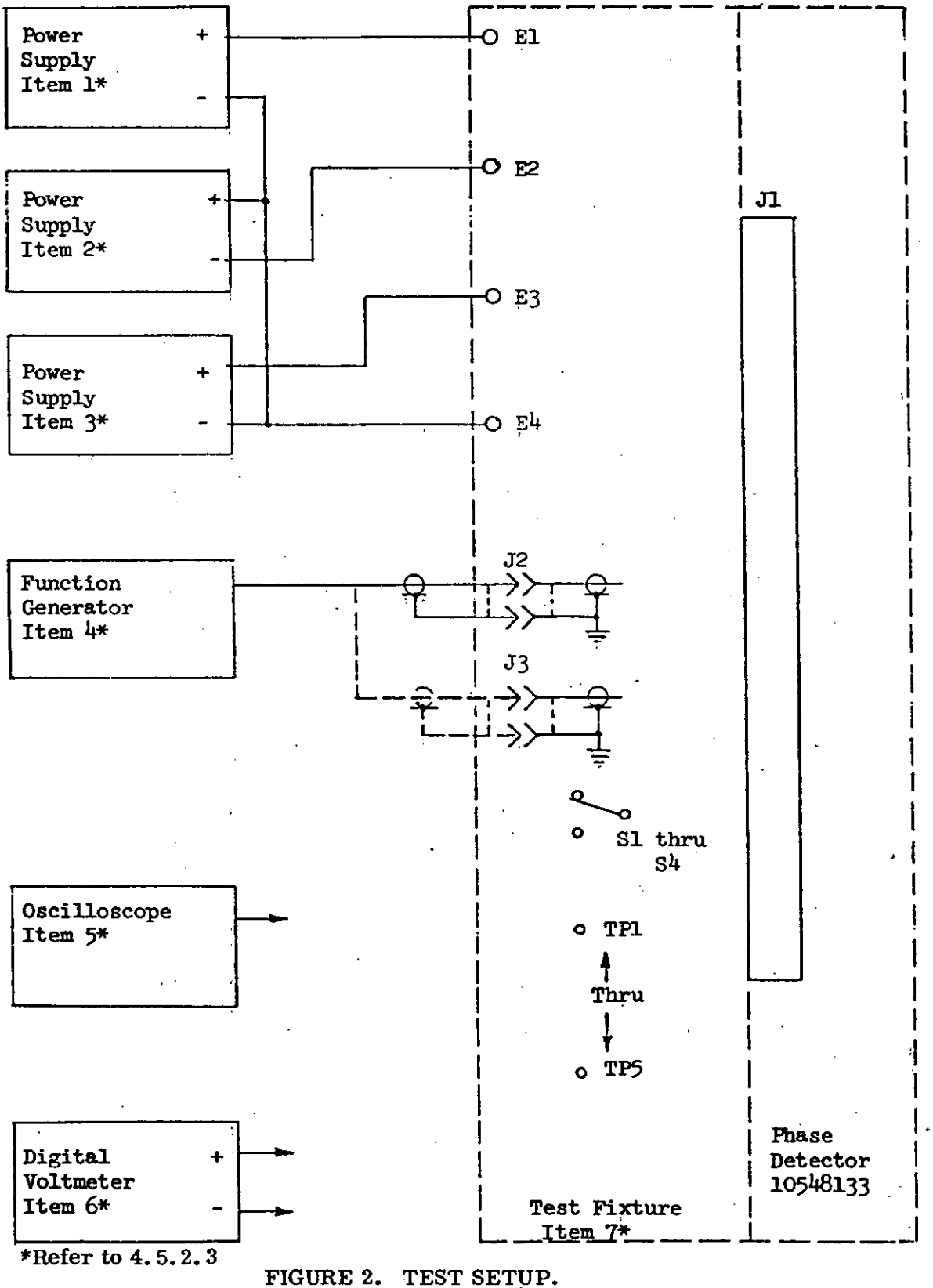
4.6.3.3 Sum signal strength response.

4.6.3.3.1 ECM relay driver signal output. - With the input signal conditions as specified in 3.5.3.1 measure the ECM relay driver output signal at TP2, on the test fixture, item 7, as specified in Table III. Use R20 on the phase detector board assembly to adjust for signal output conditions of Table III, step 1. Use switch S3 on the test fixture to supply logic levels as specified in Table III. Switch S3 supplies a logic 1 in position 1 and a logic 0 in position 2.

4.6.3.3.2 Target output signal. - Use R22 on the phase detector board assembly to adjust for signal output conditions as specified in 3.5.3.2. Observe that output signal at TP4 on the test fixture, item 7, is as specified in 3.5.3.2.

4.6.3.4 Track relay driver signal output. - Using switch S4 on the test fixture to apply logic levels as specified in steps 1 and 2 or Table IV, observe that the track relay driver signal output at TP3 on the test fixture is as specified in Table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:32Z
Check the source to verify that this is the current version before use.



*Refer to 4.5.2.3

FIGURE 2. TEST SETUP.

Source: https://assist.dla.mil - Downloaded: 2026-10-22T20:32Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. - Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10548133. The level of protection shall be specified in the procurement document.

6. NOTES

6.1 Intended use. - The phase detector assembly is used in Range Computer CP-888/VPS-2, part of Radar Set AN/VPS-2. The phase detector assembly converts doppler amplifier output sum and difference signals to digital signals and produces a digital error signal in accordance with their phase relationship.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable packaging data sheet number (see 5.1).

c. Selection of applicable level of preservation packaging and packing.

d. Level of protection required.

e. Applicable stock number.

f. Requirement for first article submission.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1285-A250

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:32Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:32Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
Department of the Army

OFFICIAL BUSINESS

Commanding Officer
Frankford Arsenal
ATTN: SMUFA-T1000
Philadelphia, Pa. 19137

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:32Z
Check the source to verify that this is the current version before use.